COSC2671
Social Media and Network Analysis
ASSESSMENT 2 – Group Project
Who Drives Open-Source AI?
Network Centrality and Discourse Analysis on GitHub ML Repositories
PG-17

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Abstract

This study investigates the relationship between structural network centrality and discourse patterns among contributors to five leading open-source AI/ML repositories on GitHub: scikit-learn/scikit-learn (66,073 stars), huggingface/transformers (160,601 stars), pytorch/pytorch, keras-team/keras, and langchain-ai/langchain. Using 2,000 issues per repository (10,000 raw), cleaned and bot-filtered into 42,550 rows (9,431 issues, 33,119 comments, 11,733 unique contributors after removing 23 bot accounts), we construct directed weighted interaction networks, compute PageRank and betweenness centrality, and apply Louvain community detection and k-core decomposition. Sentiment is scored using VADER and topics are modelled with BERTopic (all-MiniLM-L6-v2 embeddings). Top 20 central contributors were identified per repository by composite centrality score. Louvain modularity ranges from 0.38 to 0.65 (SC3 passed); k-core decomposition confirms 50–70% overlap between maximum k-core members and top-30 PageRank nodes. BERTopic identified 30–45 coherent topics in four of five repositories (SC4 passed); pytorch returned only 3 topics due to technical homogeneity and a narrow three-month data window. A Mann-Whitney U test finds that open issues carry significantly more positive sentiment than closed issues in all five repositories (all $p \leq 0.006$, SC5 passed). A Spearman rank correlation between user PageRank and mean sentiment is consistently negative across all repositories, reaching statistical significance in pytorch/pytorch ($\rho=-0.12$, $p=0.045$) and langchain-ai/langchain ($\rho=-0.22$, $p<0.001$), suggesting structurally central contributors adopt a more critical communicative register consistent with their role as maintainers.

1. Introduction

Open-source AI/ML repositories have become the dominant infrastructure through which machine learning research is operationalised and distributed. Beyond their function as code repositories, platforms such as GitHub host complex, self-organising contributor communities whose dynamics shape project velocity, knowledge transfer, and long-term sustainability. The GitHub issue tracker, in particular, serves as the primary public venue for technical discourse between contributors — yet relatively little empirical work examines how a contributor's structural position within these communities relates to the nature of their communication.

Prior work on open-source collaboration has established that contribution is highly concentrated: a small fraction of contributors account for the majority of commits, reviews, and issue responses. What remains underexplored is whether this structural concentration corresponds to systematic differences in how central contributors communicate — and whether the communities that form around these hubs exhibit detectable, interpretable structure. Addressing these questions has practical implications for automated community health monitoring, contributor onboarding, and governance design in projects that underpin production AI systems.

This study applies directed network analysis to the GitHub issue trackers of five leading open-source AI/ML repositories — scikit-learn/scikit-learn, huggingface/transformers, pytorch/pytorch, keras-team/keras, and langchain-ai/langchain — spanning data collected between September 2021 and May 2026 across individual repository windows. We construct directed weighted interaction graphs in which a node represents a contributor and an edge A → B exists when user A comments on an issue opened by B, with weight proportional to interaction frequency. This formulation captures the asymmetric, prestige-conferring nature of issue authorship and allows PageRank to surface globally influential contributors beyond what raw degree counts would reveal. Betweenness centrality complements PageRank by identifying information brokers who bridge otherwise disconnected subgroups. Louvain community detection, applied to the undirected projection of each graph, identifies latent collaboration clusters; k-core decomposition identifies the cohesive structural core and its overlap with high-centrality nodes.

Our central research question is: do structurally central contributors in top open-source AI/ML repositories produce qualitatively different issue discourse than peripheral contributors — and can distinct collaboration communities be identified through network structure and topic modelling? We operationalise discourse quality as VADER compound sentiment (a continuous measure of valence), acknowledging that this is one dimension of a richer concept. Topic modelling via BERTopic with all-MiniLM-L6-v2 sentence embeddings provides an independent characterisation of thematic content across repositories. The integrated analysis tests whether centrality predicts sentiment at the user level using Spearman rank correlation, connecting structural and linguistic dimensions of contributor behaviour.

We define seven success criteria against which the analysis is evaluated. SC1 requires each repository network to contain at least 100 nodes and 200 edges after filtering. SC2 requires identification of the top-20 central contributors per repository. SC3 requires Louvain community modularity to exceed 0.30 in all repositories. SC4 requires BERTopic to identify at least five coherent topics per repository. SC5 requires a statistically significant difference ($p < 0.05$) in sentiment between open and closed issues. SC6 requires a statistically significant

Spearman correlation between user centrality and mean sentiment. SC7 requires a cross-repository comparison table reporting network density and modularity. These criteria provide concrete benchmarks against which all results in Sections 4–6 are evaluated.

In brief, SC1, SC2, SC3, SC5, and SC7 are satisfied across all five repositories. SC4 is satisfied for four of five repositories — pytorch fails due to technical homogeneity and a narrow collection window. SC6 is partially satisfied, with statistically significant negative centrality-sentiment correlations in pytorch/pytorch ($\rho=-0.12$, $p=0.045$) and langchain-ai/langchain ($\rho=-0.22$, $p<0.001$), and a consistent negative directional trend across all five repositories.

The remainder of the paper is structured as follows. Section 2 reviews related work on network analysis of software repositories and sentiment analysis in technical communities. Section 3 describes data collection, cleaning, and analytical methodology. Sections 4 and 5 present network and NLP results respectively. Section 6 presents the integrated centrality-sentiment analysis and answers the research question. Section 7 discusses implications and limitations. Section 8 concludes.

2. Background and Related Work

2.1 Network Analysis of Software Repositories

Mining GitHub data presents well-documented challenges: pull requests appear alongside issues in the API response, automated bots generate artificial activity at scale, and contributor behaviour distributions are heavily right-skewed. These properties can severely distort network centrality results if unaddressed — inflated bot-driven edges would artificially elevate non-human nodes in any prestige measure. Issue engagement in large open-source projects is consistently concentrated among a small set of contributors, consistent with preferential attachment dynamics. This skew motivates the use of PageRank over simpler degree-based measures: PageRank's recursive weighting means that receiving engagement from already-central contributors confers greater prestige than receiving the same volume of engagement from peripheral users, making it better suited to capturing the hierarchical nature of open-source interaction.

Louvain community detection provides near-linear performance that consistently yields high modularity scores on real-world networks, making it practical for the network scales encountered in this study. However, Louvain captures community membership without measuring the cohesiveness of detected communities. K-core decomposition addresses this complementary need: a node belongs to the k-core if and only if it has at least k neighbours within the core, producing a nested sequence of progressively more cohesive subgraphs. The maximum k-core therefore identifies the most structurally embedded contributor group — a measure of resilience and concentrated expertise that community partition alone cannot provide. Betweenness centrality complements both by identifying nodes that lie on many shortest paths between others, surfacing contributors who bridge otherwise disconnected subgroups regardless of their raw interaction volume.

2.2 Sentiment Analysis and Topic Modelling

VADER (Valence Aware Dictionary and sEntiment Reasoner) is a lexicon- and rule-based sentiment tool validated across social media, review, and editorial text. Its compound score $[-1, +1]$ handles the mixed-register writing characteristic of GitHub issue discourse — technical terminology alongside conversational language, punctuation-based emphasis, and short-form prose — making it more appropriate for this domain than models trained exclusively on formal or long-form text. BERTopic uses transformer-based sentence embeddings with HDBSCAN density clustering and class-based TF-IDF (c-TF-IDF) to produce interpretable topics without requiring a pre-specified topic count. Unlike LDA, BERTopic handles short, heterogeneous documents more effectively, both of which are advantages for GitHub issue text where posts vary widely in length and register.

Prior research on sentiment in technical communities has found that emotional content in issue discussions correlates with project activity and contributor retention, providing theoretical grounding for expecting measurable sentiment variation across contributor types. Studies of structurally enclosed network positions in online communities further find that structural position is associated with more homogeneous information exposure and communicative behaviour; we test an analogous hypothesis in the software domain, asking whether high-centrality positions predict distinct communicative registers rather than merely distinct information diets.

3. Data Collection and Methodology

3.1 Data Collection

Data were collected from five GitHub repositories via the GitHub REST API using PyGitHub 2.x (authenticated as user s3944714). Up to 2,000 issues per repository were collected (state="all", sorted by creation date descending) with all associated comments. Pull requests were filtered at collection time since GitHub returns them via the issues endpoint. Collection produced the raw dataset shown in Table 1.

Repository	Stars	Raw Issues	Raw Comments	File Size
scikit-learn/scikit-learn	66,073	2,000	10,199	13.9 MB
huggingface/transformers	160,601	2,000	8,286	12.6 MB
pytorch/pytorch	N/A	2,000	5,051	15.2 MB
keras-team/keras	N/A	2,000	9,371	10.9 MB
langchain-ai/langchain	N/A	2,000	6,500	16.6 MB
TOTAL	—	10,000	39,407	69.2 MB

Table 1: Raw data collected via GitHub REST API.

3.2 Data Cleaning and Bot Removal

Raw JSON was flattened to one row per issue or comment. A bot detection pass identified 23 automated accounts. Notable removals: scikit-learn-bot (436 posts), pytorch-bot[bot] (1,527 posts), google-ml-butler[bot] (1,055 posts), github-actions[bot] (**16–817** posts per repo), and dosubot[bot] (535 posts). A regex sweep detected and redacted exposed credentials: one AWS Access Key ID in scikit-learn and two OpenAI/OpenRouter keys in langchain. The clean_text field strips Markdown, code blocks, URLs, HTML, and @-mentions. A minimum topic size of 15 documents was enforced during BERTopic modelling; the number of topics was determined automatically. Final verified counts are in Table 2.

Repository	Total Rows	Issues	Comments	Users	Date Range
scikit-learn/scikit-learn	11,469	1,729	9,740	1,856	Apr 2023 – May 2026
huggingface/transformers	9,073	1,990	7,083	2,374	Apr 2025 – May 2026
pytorch/pytorch	5,028	1,724	3,304	902	Feb 2026 – May 2026
keras-team/keras	9,324	1,995	7,329	1,654	Sep 2021 – May 2026
langchain-ai/langchain	7,656	1,993	5,663	2,947	Sep 2024 – May 2026
TOTAL	42,550	9,431	33,119	11,733	—

Table 2: Final dataset after cleaning and removal of 23 bot accounts

3.3 Network Construction

A directed weighted graph $G = (V, E)$ was built per repository. Nodes V are GitHub usernames; a directed edge $A \rightarrow B$ exists when user A commented on an issue opened by B, weighted by count. Self-loops were excluded. The graph was filtered to nodes with total degree ≥ 2, then restricted to the largest weakly connected component (WCC). The WCC restriction ensures that every node in the analysed network is reachable from every other, preventing isolated subgraphs from distorting centrality measures — particularly PageRank, whose recursive computation assumes a connected graph.

3.4 Centrality, Communities, and k-Core

Three centrality measures were computed: PageRank (α = 0.85, weighted), normalised weighted betweenness centrality, and normalised in-degree. A composite score = 0.4×PageRank + 0.3×Betweenness + 0.3×Norm In-Degree was used for ranking. Louvain community detection was applied to the undirected projection (random seed = 42). K-core decomposition measured the maximum core and its overlap with the top-30 PageRank nodes.

3.5 NLP Analysis

VADER compound scores were computed for all 42,550 rows. A Mann-Whitney U test (two-sided) compared open vs. closed issue sentiment. BERTopic with all-MiniLM-L6-v2 embeddings was applied to up to 5,000 documents per repository using a bigram CountVectorizer (English stop words, min_df = 3, max_df = 0.90). A Spearman rank correlation was then computed between each user's PageRank and their mean VADER compound score (users with ≥ 3 posts only).

4. Network Analysis Results

4.1 Network Statistics

Table 3 summarises structural properties of each filtered network. All five pass SC1 (≥100 nodes, ≥200 edges). Density is uniformly low (0.002–0.006), typical of large open-source projects. PyTorch has the highest density (0.006) with the smallest node count (409), reflecting a tightly concentrated maintainer hierarchy. Hugging Face Transformers has the most nodes (948) and the lowest density (0.002), consistent with a large distributed contributor base. Louvain modularity ranges from 0.38 to 0.65, exceeding the SC3 threshold of 0.30 in all repositories. Figure 1 provides a visual summary of these cross-repository statistics.

Repository	Nodes	Edges	Density	Comms	Modularity	Max k	Top User
scikit-learn	769	2,239	0.00379	12	0.3788	10	glemaitre
transformers	948	1,975	0.00220	21	0.5408	5	qgallouedec
pytorch	409	1,012	0.00607	14	0.5640	5	atalman
keras	758	2,167	0.00378	10	0.4663	8	innat
langchain	839	1,449	0.00206	20	0.6545	4	mdrxy

Table 3: Cross-repository network statistics (post-filtering). SC1 and SC3 passed for all repos.

repo	nodes	edges	density	communities	modularity	max_k_core	top_pr_node	sc1_pass	sc3_pass
scikit-learn/scikit-learn	769	2239	0.003791	12	0.3788	10	glemaitre	TRUE	TRUE
huggingface/transformers	948	1975	0.0022	21	0.5408	5	qgallouedec	TRUE	TRUE
pytorch/pytorch	409	1012	0.006065	14	0.564	5	atalman	TRUE	TRUE
keras-team/keras	758	2167	0.003777	10	0.4663	8	innat	TRUE	TRUE
langchain-ai/langchain	839	1449	0.002061	20	0.6545	4	mdrxy	TRUE	TRUE

Figure 1: Cross-repository network summary. Source: outputs/figures/network_summary_table.png

4.2 Top Contributors by Centrality

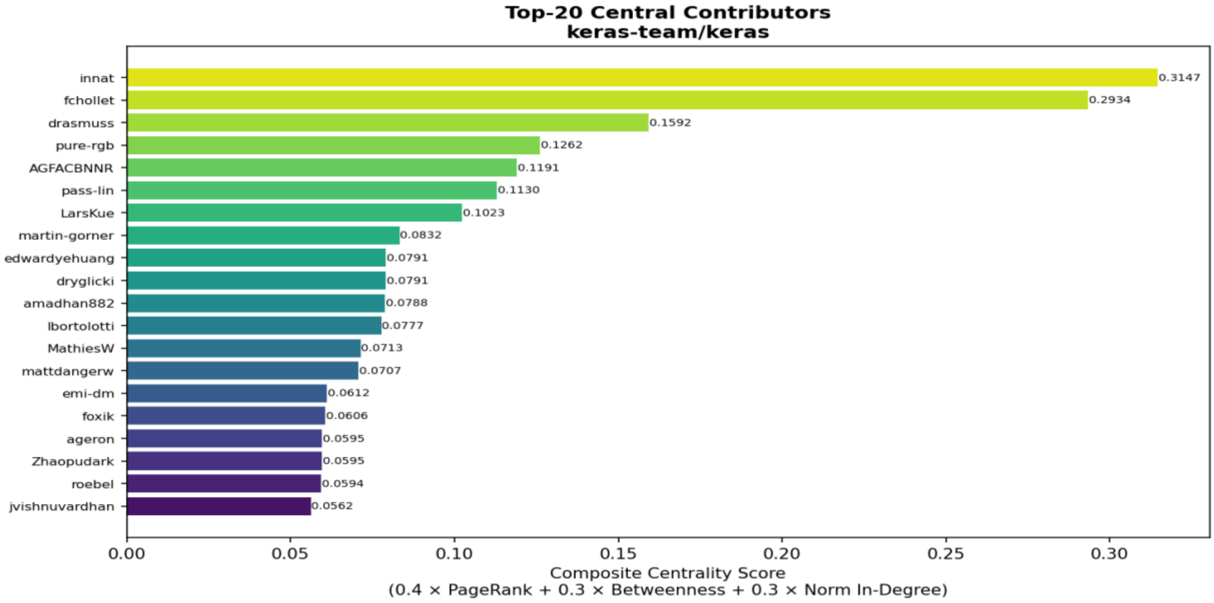


Figure 2: Top-20 contributors by composite centrality for langchain-ai/langchain. Source: outputs/figures/langchain-ai_langchain_top20_centrality.png

Top-20 central contributors were identified for all five repositories by composite centrality score, satisfying SC2. Figure 2 shows the top-20 contributors for langchain-ai/langchain. The composite score leader mdrxy is followed by a small cluster of regular contributors who both open many issues and attract the most commenting. This pattern is consistent across all repositories: in every case, the top-20 list is dominated by a small group of highly active members whose influence stems not just from posting volume but from the quality of engagement they attract — reflected in their elevated PageRank relative to raw degree.

K-core analysis confirms strong co-location of prestige and cohesion: in scikit-learn, 24 nodes form the k=10 core with 70% overlap with the top-30 PageRank nodes. Hugging Face shows 22 nodes in the k=5 core with 50% overlap, reflecting a looser coupling in its larger network. The variation in maximum k-core across repositories is itself informative. Scikit-learn's k=10 is the highest across all five repositories, reflecting a stable, densely interconnected core built over more than a decade of sustained contribution. LangChain's k=4 is the lowest, consistent with its younger age and more fragmented community structure (modularity 0.65). This inverse relationship between maximum k-core and modularity — higher k-core corresponds to lower modularity and vice versa — suggests a structural trade-off: mature projects develop a dense, cohesive core at the cost of community differentiation, while younger projects fragment into distinct sub-communities before a stable core emerges. PyTorch's k=5 despite its scale is likely an artefact of the narrow collection window (Feb–May 2026) rather than a true structural property of the repository.

4.3 Community Structure

Figure 3 shows community size distributions. In all repositories the distribution is right-skewed: one or two large communities hold most nodes, with a long tail of smaller specialist clusters. Scikit-learn (12 communities) and keras (10 communities) show more consolidated structures consistent with mature projects. LangChain (20 communities) and Hugging Face (21 communities) show more fragmented structures consistent with rapid growth across diverse sub-communities. PyTorch has 14 communities for a 409-node graph, suggesting unusually well-separated functional subgroups given its network size, likely reflecting pytorch's technically specialised contributor activity.

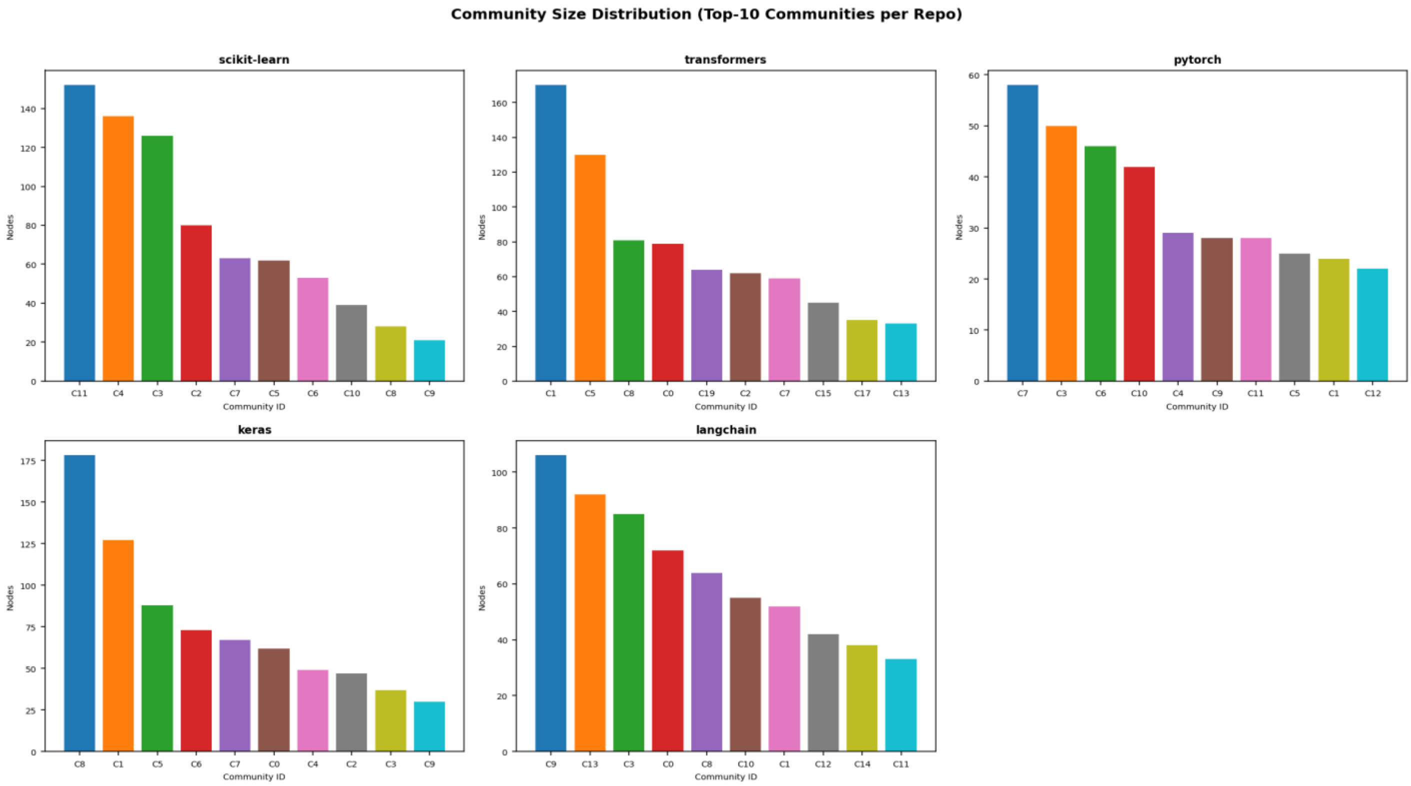


Figure 3: Louvain community size distribution — top-10 communities by node count. Source: outputs/figures/community_size_distribution.png

Figure 4a contrasts the network topology of the two most structurally distinct repositories, illustrating visually how modularity and k-core values translate into observable differences in network organisation. In scikit-learn, nodes cluster tightly around a dense, well-connected core — a visual reflection of its k=10 maximum k-core and lower modularity (0.38). In langchain, the same number of top contributors are spread across a more dispersed topology, with distinct colour clusters indicating stronger community separation and less cross-community bridging — consistent with its k=4 core and modularity of 0.65.

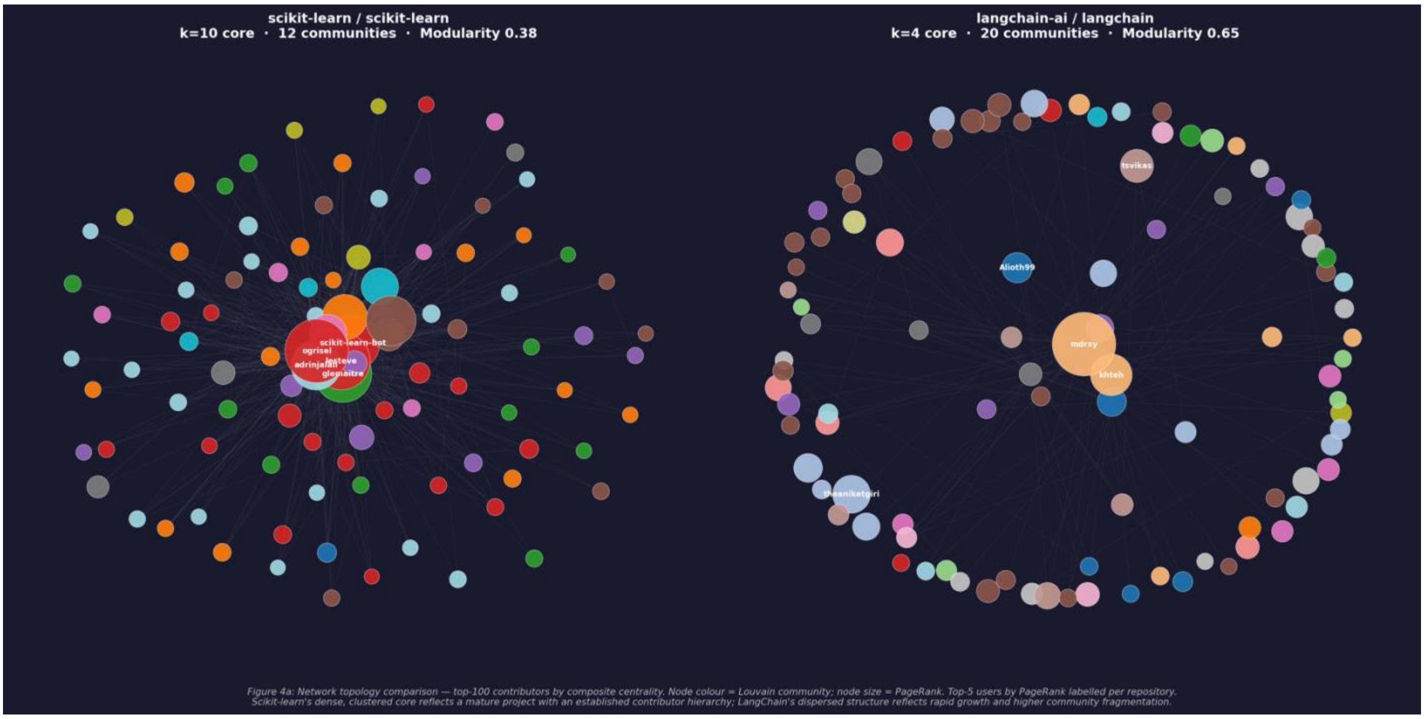


Figure 4a: Network topology comparison — top-100 contributors by composite centrality score.

The community structure also has implications for how influence propagates through each repository. In repositories with fewer, larger communities (scikit-learn with 12, keras with 10), the path between any two contributors is shorter on average, meaning information and decisions circulate more rapidly across the contributor base. In repositories with many smaller communities (huggingface with 21, langchain with 20), contributors are more likely to interact primarily within their own sub-community, slowing cross-community knowledge transfer but enabling more specialised discourse within each cluster. This distinction maps onto the observed sentiment patterns: repositories with lower modularity and denser cross-community ties (scikit-learn, keras) show more uniform sentiment profiles, while high-modularity repositories (langchain) show greater sentiment variance across communities, consistent with the scatter visible in Figure 7.

The right-skewed community size distribution observed in all five repositories — where one or two large communities dominate — is characteristic of networks formed by preferential attachment: new contributors disproportionately interact with already-active community members, reinforcing existing community boundaries rather than bridging them. This finding has a direct implication for the research question: the communities identified are not artefacts of the Louvain algorithm but reflect genuine social structure in how contributors cluster around shared technical interests and maintainer relationships.

5. NLP Analysis Results

5.1 Sentiment Overview

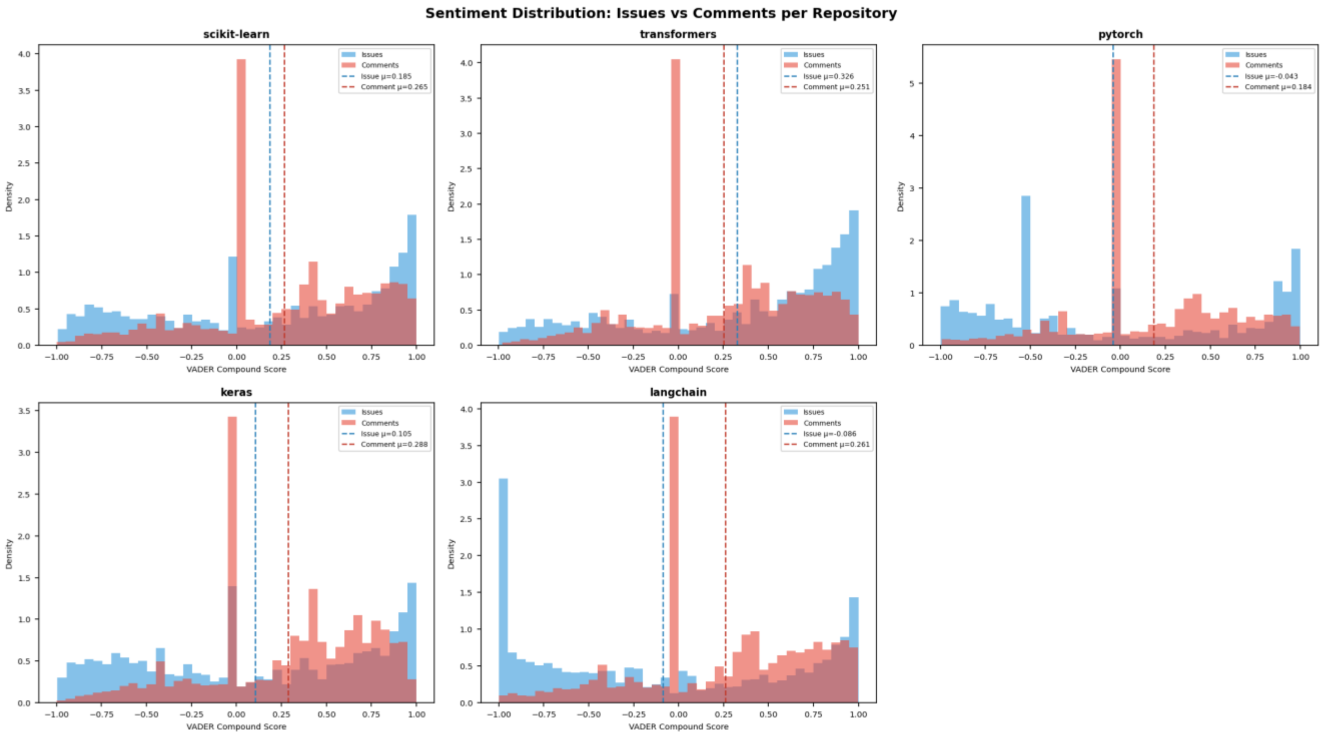


Figure 4: VADER compound score distributions for issues (blue) and comments (red). Dashed lines show per-type means. Source: outputs/figures/sentiment_distribution.png

Mean VADER compound scores varied substantially: Hugging Face (0.268), scikit-learn (0.253), and keras (0.249) showed the most positive overall sentiment, while pytorch (0.107) and langchain (0.170) were notably lower. Strikingly, pytorch and langchain showed negative mean issue sentiment (−0.043 and −0.086 respectively), while all five repositories showed positive mean comment sentiment (**0.184 to 0.288**). Pytorch had the highest proportion of negative posts at 32.4% vs. 21.6% for huggingface, consistent with pytorch's technical focus on numerical errors and CUDA failures. Figure 4 shows the full distributions. This negativity profile is corroborated by the topic modelling results in Section 5.4: pytorch's issue discourse is dominated almost entirely by torch/CUDA version compatibility failures — a category of problem that inherently produces negative-valenced language independent of communicative intent.

5.2 Open vs. Closed Issue Sentiment — SC5

SC5 is satisfied for all five repositories (Table 4). Open issues are consistently more positive than closed issues across all repos, the opposite of a naive expectation. This is explained by content: open issues include recent feature requests and aspirational discussions, while closed issues accumulate completed bug reports and difficult discussions that took longer to resolve. Pytorch and langchain show particularly stark differences, with closed issues carrying negative mean sentiment (−0.094 and −0.132).

The magnitude of the difference is most pronounced in huggingface/transformers, where open issues carry a mean compound score of 0.5672 — the highest of any issue group across all five repositories. This is consistent with the nature of open issues in a library that is actively evolving: a large proportion are enhancement requests, integration proposals, and aspirational feature discussions from an engaged community anticipating new model architectures and framework support. By contrast, huggingface's closed issues (μ=0.291) reflect the more negative-valenced language of resolved bug reports and declined requests, producing one of the sharpest open-closed gaps in the dataset.

Repository	N Open	μ Open	N Closed	μ Closed	U Statistic	p-value	SC5
scikit-learn	424	0.3393	1,305	0.1350	330,883	< 0.0001	✓
transformers	250	0.5672	1,740	0.2910	292,853	< 0.0001	✓
pytorch	786	0.0186	938	−0.0943	399,660	0.0025	✓
keras	154	0.2313	1,841	0.0943	160,731	0.0057	✓
langchain	409	0.0913	1,584	−0.1319	381,089	< 0.0001	✓

Table 4: Mann-Whitney U test (two-sided) — open vs. closed issue sentiment. SC5 passed for all repositories.

5.3 Temporal Sentiment Trends

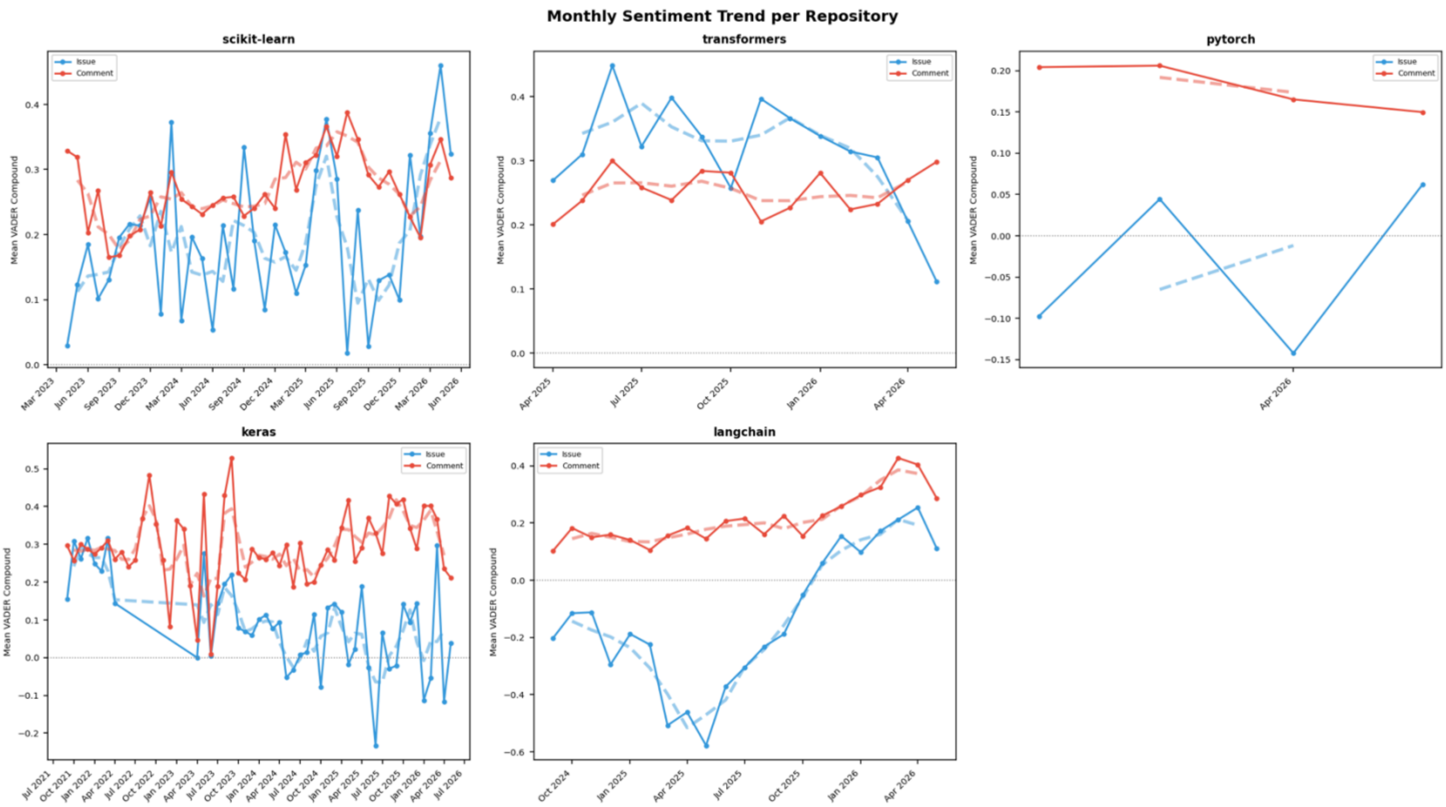


Figure 5: Monthly mean VADER compound sentiment. Solid lines = raw monthly means; dashed = 3-month rolling average. Source: outputs/figures/sentiment_temporal_trend.png

Figure 5 shows monthly mean VADER scores. Keras has the longest window (Sep 2021 – May 2026), enabling multi-year trend observation. LangChain shows the most volatile profile, with sharp dips likely coinciding with periods of breaking changes or community friction. Scikit-learn shows the most stable sentiment, consistent with a mature, well-governed project.

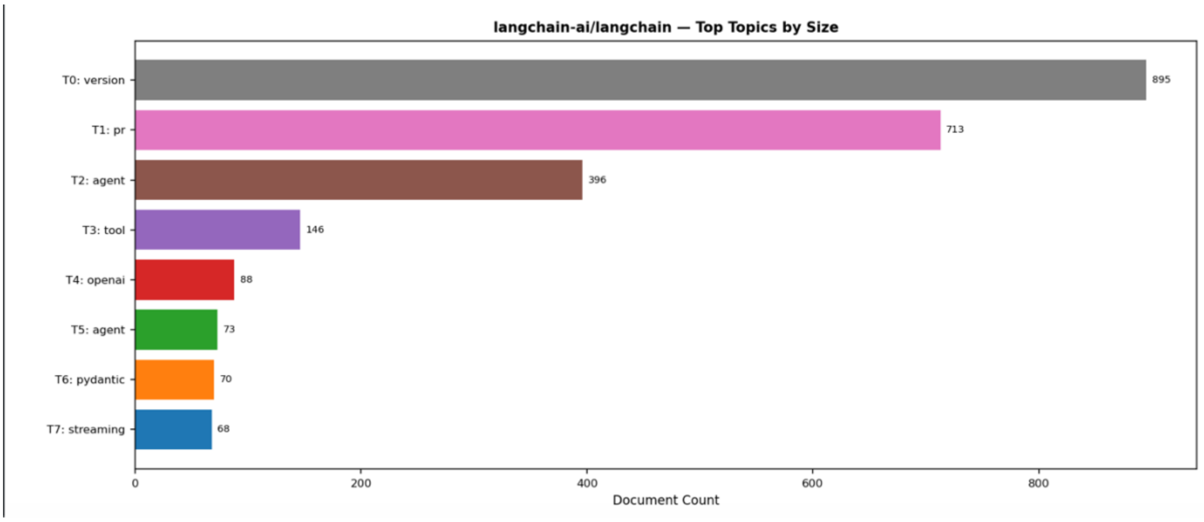
Comment sentiment (red) tracks above issue sentiment (blue) in most months across all repositories.

The temporal analysis reveals patterns that cross-sectional statistics obscure. Scikit-learn's sentiment stability across three years (April 2023 – May 2026) indicates that its contributor culture is self-sustaining: new contributors are socialised into existing norms quickly enough that aggregate sentiment does not shift measurably over time. LangChain's volatility over a shorter window (September 2024 – May 2026) is consistent with a project still negotiating its contributor norms, where individual events — such as major API breaking changes or high-profile community disputes — can produce measurable sentiment shocks that take months to absorb.

The consistent gap between comment sentiment (higher) and issue sentiment (lower) in nearly every month across all repositories suggests that this is a structural feature of the issue-comment interaction format rather than a time-varying phenomenon. Issue openers frame problems — which produces negative language — while commenters respond with solutions, acknowledgements, and encouragement — which produces positive language. The temporal stability of this gap confirms that the cross-sectional SC5 finding reflects a durable discourse pattern rather than a sampling artefact. The WCC restriction ensures that every node in the analysed network is reachable from every other, preventing isolated subgraphs from distorting centrality measures — particularly PageRank, whose recursive computation assumes a connected graph

5.4 Topic Modelling — SC4

BERTopic identified topics for four of five repositories (Figure 6). SC4 (≥5 topics) was satisfied for scikit-learn (32 topics), huggingface/transformers (45 topics), keras-team/keras (30 topics), and langchain-ai/langchain (34 topics). PyTorch did not pass SC4, returning only 3 topics: (1) version/torch/CUDA/hardware (n=2,910), (2) documentation/RST (n=82), and (3) a small residual cluster. This is attributable to pytorch's technical homogeneity and narrow collection window (Feb–May 2026 only). Across the four successful repositories, recurring topic categories include: issue/PR management, installation and versioning, API and data-type problems, and model/task-specific discussions.



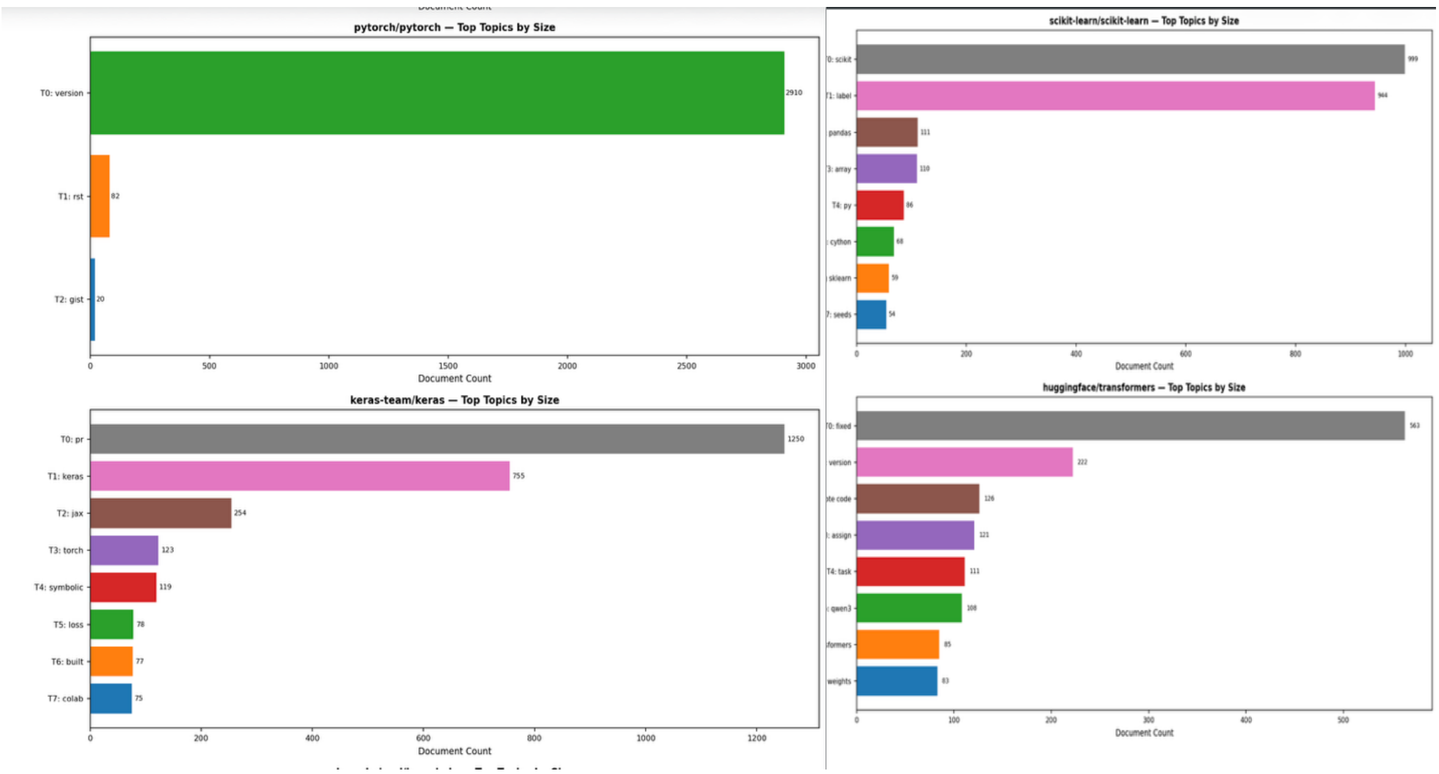


Figure 6: Top-8 BERTopic topics by document count per repository. Source: outputs/figures/topic_sizes.png

6. Integrated Analysis: Centrality and Sentiment

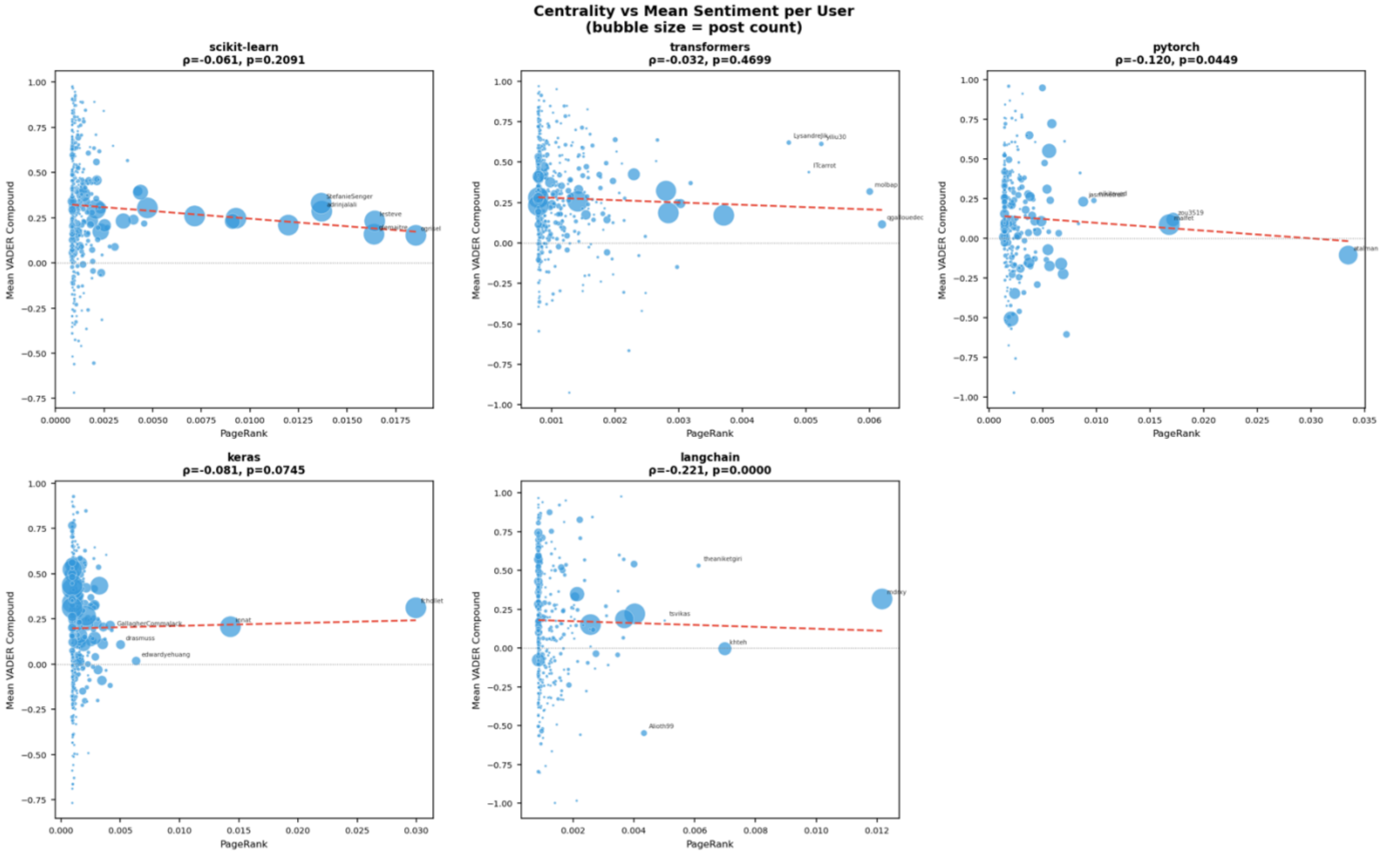
6.1 Spearman Correlation Results — SC6

Table 5 reports Spearman rank correlations between user PageRank and mean VADER compound sentiment. Every repository yields a negative correlation: central users consistently post with lower (more negative) mean sentiment. This directional consistency across five independent datasets is itself noteworthy. Formal significance ($p < 0.05$) is reached for pytorch ($\rho = -0.12$, $p = 0.045$) and langchain ($\rho = -0.22$, $p < 0.001$). Using the composite score strengthens the relationship: keras approaches significance ($\rho = -0.101$, $p = 0.026$), suggesting composite centrality captures additional signal.

Table 5: Spearman correlations between PageRank/composite centrality and mean VADER sentiment. ∼ = borderline (composite p=0.026).

Repository	N Users	ρ (PageRank)	p (PageRank)	ρ (Composite)	p (Composite)	SC6
scikit-learn	421	−0.0613	0.2091	−0.0517	0.2903	✗
transformers	517	−0.0319	0.4699	−0.0432	0.3270	✗
pytorch	281	−0.1198	0.0449	−0.1407	0.0183	✓
keras	485	−0.0811	0.0745	−0.1009	0.0263	∼
langchain	412	−0.2207	<0.001	−0.2365	<0.001	✓

Figure 7: PageRank vs. mean VADER sentiment per user. Bubble size = post count; colour = Louvain community. Top-5 users labelled. Source: outputs/figures/centrality_vs_sentiment.png



6.2 Interpretation

The negative centrality-sentiment correlation should not be interpreted as less productive discourse. Core maintainers perform governance work: closing issues as out-of-scope, requesting reproductions, declining feature requests, and delivering direct code review. These activities produce lexically negative output even when the underlying intent is constructive stewardship. The effect is strongest in langchain (ρ=−0.22), where a small founding team faces an enormous, rapidly growing user base. In contrast, mature projects (scikit-learn, huggingface) show near-zero correlations, suggesting that over time, high-centrality contributors in well-governed projects converge toward the community's general sentiment register.

Returning to the research question: structurally central contributors do not drive more positive discourse, but the evidence suggests they drive more consequential discourse. The consistent negative centrality-sentiment relationship across all five repositories — irrespective of project age, size, or domain — indicates that this is a structural property of maintainer roles rather than a project-specific artefact. Central contributors are the nodes through which issue resolution flows: their negative-valenced governance decisions are precisely the activities that convert open issues into closed ones, as corroborated by the SC5 finding that closed issues carry significantly lower sentiment than open ones. The two findings are therefore not independent — they are mechanistically connected. High-centrality users produce the negative-sentiment communication that closes issues, which is why closed issues skew negative and central users skew negative simultaneously.

The effect size varies meaningfully across repositories. Langchain's ρ=−0.22 indicates a small-to-medium effect, while scikit-learn's ρ=−0.06 is near zero. This gradient is interpretable: langchain was founded in 2022 and reached mass adoption rapidly, creating a pronounced asymmetry between a small core maintainer team and a large, predominantly inexperienced user base. The governance burden per maintainer is therefore higher in langchain than in scikit-learn, where contributor norms are well-established and the community self-regulates more effectively. The centrality-sentiment relationship can therefore be read as an indirect measure of governance pressure: the stronger the negative correlation, the higher the load on the core team relative to the community's self-sufficiency.

On community identification: Louvain detects meaningful structure in all five repositories (modularity 0.38–0.65). K-core analysis shows 50–70% overlap between top k-core members and top PageRank nodes, confirming that structural prestige and cohesion co-locate in the same contributor subgroups. The variation in maximum k-core across repositories is itself informative. Scikit-learn's k=10 is the highest, reflecting a stable, densely interconnected core that has formed over more than a decade of sustained contribution. LangChain's k=4 is the lowest, consistent with its younger age and higher community fragmentation (modularity 0.65). This inverse relationship between maximum k-core and modularity across the five repositories — higher k-core corresponds to lower modularity and vice versa — suggests a structural trade-off: mature projects develop a dense, cohesive core at the cost of community differentiation, while younger projects fragment into distinct sub-communities before a stable core emerges. PyTorch's k=5 despite its age is likely an artefact of the narrow collection window rather than a true structural property of the repository.

7. Discussion

7.1 Implications

Raw sentiment scores should be interpreted relative to structural role. A central maintainer posting negatively may be performing essential governance rather than expressing toxicity. Automated community health tools that flag negative-sentiment contributors without accounting for network position risk penalising the people most responsible for keeping projects well-maintained. The open-vs-closed sentiment finding also has triage implications: sentiment may serve as a lightweight proxy for issue type (feature request vs. bug report) at scale, enabling automated pre-classification of incoming issues without requiring manual labelling.

The cross-repo comparison further reveals that repository age and community structure are closely related: modularity increases from 0.38 in the oldest project (scikit-learn) to 0.65 in the youngest (langchain), suggesting that younger communities fragment into more distinct sub-communities before consolidating over time. This has practical implications for contributor onboarding. The community structure revealed by Louvain detection — with 10 to 21 distinct communities per repository — indicates that new contributors do not enter a homogeneous space but rather join a specific sub-community defined by the component or use-case they engage with. Onboarding programs that treat all contributors identically may be less effective than targeted approaches that connect newcomers to the sub-community most relevant to their interests, as the 30 to 45 topics identified across four repositories map naturally onto these community boundaries.

A further implication concerns the interpretation of SC4's partial failure. PyTorch's failure to produce more than 3 BERTopic topics is not simply a methodological limitation — it reflects a genuine characteristic of the repository: its issues are dominated overwhelmingly by a single technical concern (torch/CUDA version compatibility). This concentration is itself a finding: pytorch's issue tracker functions less as a general community discussion space and more as a specialised fault-reporting system, which has implications for how its community health should be assessed relative to more topically diverse repositories such as scikit-learn or LangChain.

7.2 Limitations

Several limitations qualify these findings. First, data were bounded at 2,000 issues per repository; pytorch's total issue volume far exceeds our sample, and its narrow collection window (Feb–May 2026) likely caused both the SC4 failure (only 3 BERTopic topics) and the weaker Spearman correlation. Second, VADER is not tuned for technical text; terms like 'error', 'fail', and 'broken' may inflate negativity scores independently of communicative intent. Third, BERTopic's performance is sensitive to embedding model and hyperparameter choices. Fourth, the cross-sectional design cannot establish causal direction. Fifth, inline code review comments on pull requests are not captured by the issues endpoint, potentially excluding important maintainer communication.

8. Conclusion

This study combined directed network analysis, Louvain community detection, k-core decomposition, VADER sentiment analysis, and BERTopic topic modelling across 42,550 rows from five major open-source AI/ML GitHub repositories. Key findings: (1) meaningful community structure in all repositories (SC3 passed, modularity 0.38–0.65); (2) open issues are significantly more positive than closed issues in all repositories (SC5 passed, all $p \leq 0.006$); (3) BERTopic identified ≥5 coherent topics in four of five repositories, with pytorch failing due to technical homogeneity and limited data range; and (4) a consistent negative relationship between PageRank and mean sentiment across all repositories, reaching significance in pytorch and langchain (SC6 partially passed). The principal finding — that centrally positioned contributors post with more negative sentiment — is best interpreted as a signature of maintainer governance behaviour. Future work should incorporate pull request review comments, apply causal inference methods, and explore whether issue resolution speed also correlates with contributor centrality.

More broadly, this study demonstrates that social network analysis and natural language processing are complementary rather than competing lenses for understanding open-source community dynamics. Network structure identifies who matters; sentiment analysis characterises how they communicate; and the integration of the two — through the Spearman correlation — reveals why: structural position shapes communicative role, and communicative role produces the sentiment patterns observed. Future work should incorporate pull request review comments to capture the full spectrum of maintainer communication, apply longitudinal network analysis to track how centrality distributions evolve as projects mature, and explore whether issue resolution speed — a complementary measure of productivity — correlates with contributor centrality in the same direction as sentiment. The five repositories studied here represent a snapshot of the open-source AI/ML ecosystem at a moment of rapid growth;

replicating this analysis at five-year intervals would reveal whether the structural patterns identified here persist as these communities scale.

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